Rigorous Design Notes and Code Guide for Improving the Conditional WGAN Synthetic-Genotype Pipeline

Technical companion for `conditional_wgan_pipeline.py`

October 2026

Abstract

This document explains and formalizes a set of opt-in improvements to a conditional Wasserstein GAN with gradient penalty (cWGAN-GP) for synthetic genotype generation. It is written for a reader who is relatively new to population genetics and generative modeling. Every proposed change is defined mathematically, the failure mode it targets is stated precisely, and the corresponding implementation is identified. The revised code keeps the original PCA-space MLP as the default baseline and adds experimental alternatives rather than silently changing the scientific model. The main additions are: (i) a direct SNP-space residual 1-D convolutional WGAN; (ii) FiLM conditioning at every hidden block; (iii) optional external ancestry coefficients such as ADMIXTURE estimates; (iv) inverse-frequency population-balanced minibatch sampling; (v) PacGAN-style critic packing; (vi) multi-metric model/run selection; (vii) supplementary-figure evaluation on unphased dosages, including per-chromosome and pooled LD; (viii) checkpoint-compatible genotype sampling experiments; and (ix) separately gated membership inference and privacy diagnostics. The discussion also explains important limitations, including the distinction between unphased diploid genotype dosage and phased haplotypes, the effect of PCA truncation on covariance, and why privacy diagnostics are not differential-privacy guarantees.

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1 Code organization and reproducible experiment strategy

The implementation intentionally preserves the latest baseline pipeline and makes every new modeling idea an explicit switch. This is important scientifically: if architecture, data weighting, conditioning, and output representation are changed simultaneously, an improvement or degradation cannot be attributed to any one cause.

1.1 Files

File	Purpose
`conditional_wgan_pipeline.py`	Main run orchestration, data loading, PCA evaluation, logging, plotting, and CLI. The default backend remains the previous PCA-space model.
`cwgan_extensions/models.py`	FiLM-conditioned PCA WGAN and direct SNP-space residual Conv1D WGAN; also contains population-balanced sampling and PacGAN packing support.
`cwgan_extensions/conditioning.py`	Loader and validation for external AFR/EUR/AMR ancestry coefficients, including three-column ADMIXTURE-like `.Q` files.
`cwgan_extensions/selection.py`	Multi-metric loss-vector and composite-quality utilities.
`select_best_cwgan_run.py`	Compares completed runs using multiple fidelity metrics with robust candidate-set scaling.
`cwgan_extensions/privacy.py`	Exact-replay, real–synthetic proximity, and simple distance-based membership-inference diagnostics.
`evaluate_genotype_privacy.py`	Command-line wrapper for the pre-existing proximity privacy diagnostics.
`genotype_paper_evaluation.py`	Genotype-adapted 2023 article and supporting-figure analyses, one family per checkpoint/method.
`genotype_sampling.py`	Raw preclipping distribution plots and post-hoc 0/1/2 sampler comparison.
`evaluate_saved_genotypes.py`	Archived output discovery and checkpoint resampling/re-evaluation.
`run_membership_attacks.py`	LOGAN-style white-box critic ranking, optional black-box surrogate, with explicit train/test/condition files.

1.2 Recommended ablation order

An *ablation* is an experiment in which one component is changed while the others are held fixed. A clean progression is:

1. `baseline_pca`: reproduce the present model.
2. `pca_film`: change only how ancestry information enters the network.
3. `pca_film+balanced_sampling`: then change the minibatch population mixture.
4. `pca_film+pack_size=2`: then test critic packing.
5. Repeat the best PC-space configuration with `condition_source=external`.

6. Finally test `genotype_conv`, because this changes the representation itself and therefore is not a small perturbation of the baseline.

Use the same real validation/holdout set, seed set, SNP set, and number of generated samples when comparing configurations.

2 Population-genetic notation

Suppose there are n diploid individuals and p biallelic SNPs. The genotype dosage matrix is

$$X \in \{0, 1, 2\}^{n \times p},$$

where X_{ij} is the number of alternative alleles carried by individual i at SNP j. Thus 0 means homozygous reference, 1 heterozygous, and 2 homozygous alternative.

2.1 Allele frequency and minor-allele frequency

For SNP j, the sample alternative-allele frequency is

$$\widehat{p}_j = \frac{1}{2n} \sum_{i=1}^{n} X_{ij}.$$

The minor-allele frequency (MAF) is

$$\widehat{m}_j = \min(\widehat{p}_j, 1 - \widehat{p}_j).$$

A synthetic dataset can match a PCA cloud while still having biased $\widehat{p}_j$ or $\widehat{m}_j$, so allele-frequency fidelity must be measured separately. Synthetic-genome studies commonly compare frequency distributions and other locus-level statistics [Yelmen et al., 2021, 2023, Wharrie et al., 2023].

2.2 Heterozygosity

For an individual i, the observed heterozygosity fraction over p loci is

$$H_i = \frac{1}{p} \sum_{j=1}^{p} \mathbf{1}\{X_{ij} = 1\}.$$

This quantity is especially relevant to the current reconstruction pipeline because a continuous PCA reconstruction is converted back to discrete 0/1/2 genotypes.

2.3 Linkage disequilibrium

Linkage disequilibrium (LD) means non-random association between alleles at different loci [Slatkin, 2008]. For two biallelic loci A and B on phased haplotypes, let p_A and p_B be alternative-allele frequencies and p_{AB} the frequency of the haplotype carrying the alternative allele at both loci. Then

$$D = p_{AB} - p_A p_B, \qquad r^2 = \frac{D^2}{p_A(1 - p_A)p_B(1 - p_B)}.$$

For unphased diploid dosage data, a commonly used proxy is the squared Pearson correlation between genotype dosages at the two loci. This is not identical to phase-aware haplotype LD in every setting. Therefore the current pipeline's dosage-r^2 diagnostic should be described as a genotype-correlation LD proxy, especially when the input is unphased.

2.4 Global ancestry proportions

For $K = 3$ ancestral source groups (AFR, EUR, AMR), define

$$q_i = (q_{i1}, q_{i2}, q_{i3}), \qquad q_{ik} \geq 0, \qquad \sum_{k=1}^{3} q_{ik} = 1.$$

The vector q_i lies on the two-dimensional probability simplex Δ^2. It is the condition supplied to the generator and critic.

3 Reproducing the 2023 evaluation families on unphased data

3.1 What the papers and their code actually calculate

Yelmen et al. [2023], pp. 8–9 ("Summary statistics", "Nearest neighbour adversarial accuracy", "Nearest neighbour chain analysis", and "Membership inference attacks") specify their diagnostics. Their supplement captions (pp. 16–18, S2–S15) show which displays are repeated on 10K versus 65K SNP datasets. The earlier methods of Yelmen et al. [2021] explain LD matrices/decay, principal components and pairwise distances. In the supplied `artificial_genomes-master.zip`, inspect `plotting_notebooks/short/plot_utils.py` (`plotLDblock`, `plotPCAdensity`, `computePCAdist`, `plotPrivacyLoss`) and `plotting_notebooks/short/computeAATS.py` (`computeAAandDist`); the former performs LD and score visualizations, while the latter uses `scipy.spatial.distance.cdist(...,'cityblo` for nearest-neighbour haplotype distances. These are direct code-level precedents; they are *not* interchangeable with dosage-specific statistics. The 2023 paper explicitly states that its input has 5008 *phased* 0/1 haplotypes (p. 3, "Data"); the present input contains one *diploid* 0/1/2 dosage vector per individual.

Paper figures	Existing code reusable?	Our adaptation
Fig. 3, Fig. 5a,b; S13, S14a,b	PCA/MAF and PCA overlays: yes	Retain projection on reference PCs; additionally fit a joint PCA for paper-style PC1–PC4 real/synthetic plots and measure site-level allele frequencies.
Fig. 5c; S2	Adjacent-site LD *only partly*	Sample intrachromosomal pairs; dosage-r^2 scatter, distance decay, bin correspondence and bounded LD heatmap, both pooled and for every chromosome.
S3 and S9	No	Standardized third central moments on *diploid dosages*, not phased allele triplets.
S4 and S10	No	Window-by-window categorical dosage motifs (3^4 and 3^8 possible patterns); do not claim haplotype motifs.
S5 and S12	Existing distance utilities partly	Real–real, generated–generated and real–generated Hamming distance distributions.
Fig. 5d; S6, S14c,d	Existing AATS helper partly	Recalculate nearest-neighbour AATS in dosage Hamming space; privacy difference only with excluded subjects.
S7, S11	Existing quality metrics partly	Diagnostic radar with explicitly documented binomial-null calibration; not numerically identical to authors' historical panel.
S8	No	Fixed synthetic SNPs and corresponding real allele-frequency distribution.
Table 1	No	Nonrevisiting nearest-neighbour chains of lengths 2–5.
S15	Existing privacy utility partly	White-box critic-score ranking; opt-in black-box *surrogate* using an auxiliary PCA-space cWGAN, not necessarily the paper's exact architecture.
S1; Figs. 1, 2, 4	Not diagnostic tests	Architecture or conceptual schematics; no per-run statistic to reproduce.

3.2 Genotype LD and physical distance

Let $G_{ij} \in \{0, 1, 2\}$ denote the alternate-allele count at site j in individual i, for $i = 1, \ldots, n$ and $j = 1, \ldots, p$. Write $\bar{G}_j = n^{-1} \sum_i G_{ij}$ and $Z_{ij} = G_{ij} - \bar{G}_j$. For two polymorphic SNPs j and k, define

$$\widehat{r}^2_{jk} = \frac{(\sum_{i=1}^n Z_{ij} Z_{ik})^2}{(\sum_{i=1}^n Z_{ij}^2)(\sum_{i=1}^n Z_{ik}^2)}.$$

Here *polymorphic* means the relevant sample column has positive variance. When either denominator sum is zero the pair is excluded, never silently assigned zero LD. This phase-free genotype correlation is justified by Rogers and Huff [2009]; in admixed populations it may also reflect ancestry/population structure, so we explicitly call it *dosage correlation* rather than a unique estimate of haplotype-level D' or phased r^2. Define b_j as the genomic base-pair coordinate parsed from the name `chr:position:REF:ALT`; $d_{jk} = |b_k - b_j|$ is physical distance. *Only* pairs on the same chromosome enter any LD curve. The pooled run aggregates those pairs across chromosomes and never correlates chromosomes with one another; the per-chromosome runs use only the designated chromosome. When a coordinate cannot be parsed, the code uses SNP-order separation and labels this fallback explicitly; those curves must not be compared as base-pair decay curves.

For a physical-distance bin B consisting of sampled same-chromosome pairs, $L_{B,\mathrm{real}}$ and $L_{B,\mathrm{syn}}$ are their separate mean $\widehat{r}^2$. The paper-inspired radar coordinate is $1-\sum_B(L_{B,\mathrm{real}}-L_{B,\mathrm{syn}})^2/\sum_B L_{B,\mathrm{real}}^2$ when the denominator is nonzero. Here B ranges over nonempty log-distance bins. Binning, sample counts and the pairs themselves are saved so that this approximate calculation is auditable. Pair subsampling rather than full pair enumeration follows the scalability motivation in the 2023 paper, p. 8.

3.3 Three-point moments and genotype motifs

Define $\widehat{\sigma}_j^2 = n^{-1}\sum_i(G_{ij}-\bar{G}_j)^2$, the empirical dosage variance at SNP j, and $\widetilde{G}_{ij} = (G_{ij}-\bar{G}_j)/\widehat{\sigma}_j$ when $\widehat{\sigma}_j > 0$. For an ordered triple of sites (j,k,ℓ), the statistic plotted is

$$T_{jk\ell} = \frac{1}{n}\sum_{i=1}^{n}\widetilde{G}_{ij}\widetilde{G}_{ik}\widetilde{G}_{i\ell}.$$

Triples with zero variance at any site are excluded. We calculate equal-index-step triples for step sizes 1, 4, 16, 64, 256, 512 and 1024, and random triples, independently within chromosome. This T is a standardized *third central cross-moment* of unphased genotypes, not the authors' phased three-allele statistic. Consequently differences in absolute values between papers must not be interpreted as changed biological phase fidelity.

For motif length $k \in \{4,8\}$, a *window* is k consecutive SNPs within one chromosome, with windows nonoverlapping. Its dosage pattern is an element $\boldsymbol{g} = (g_1,\ldots,g_k) \in \{0,1,2\}^k$. The sample frequency of pattern $\boldsymbol{g}$ within window w is

$$\widehat{f}_w(\boldsymbol{g}) = \frac{1}{n}\sum_{i=1}^{n}\mathbf{1}\{(G_{i,w_1},\ldots,G_{i,w_k}) = \boldsymbol{g}\},$$

where w_t is the t-th SNP index in window w and $\mathbf{1}\{\cdot\}$ is an indicator that equals one if the event is true and zero otherwise. Since the pattern alphabet is ternary, each window has 3^k possible motifs, not the phased binary 2^k. We pool the real/synthetic window–pattern frequency pairs and report their correlation; a zero-frequency motif in both datasets is omitted to avoid inflating correlation by shared structural zeros. No haplotype phasing is inferred.

3.4 Diploid-specific quality and identity

At SNP j, let $\widehat{p}_j = (2n)^{-1}\sum_i G_{ij}$ be its alternate-allele frequency. The observed heterozygote frequency is $\widehat{h}_j = n^{-1}\sum_i \mathbf{1}\{G_{ij}=1\}$. Under Hardy–Weinberg equilibrium (HWE), a random-mating two-allele population has expected heterozygosity $h_j^{\mathrm{HWE}} = 2\widehat{p}_j(1-\widehat{p}_j)$. The diagnostic residual $\widehat{h}_j - h_j^{\mathrm{HWE}}$ is calculated in each dataset; it is descriptive and is *not* a within-population HWE hypothesis test in an admixed cohort. An exact replay is a generated row identical at every SNP to at least one real row; full-cohort replay counts use genotype-row hashes, not subsampled nearest neighbours.

Define dosage Hamming distance for individuals a,b as $d(a,b) = p^{-1}\sum_{j=1}^{p}\mathbf{1}\{G_{aj}\neq G_{bj}\}$. It measures the fraction of unequal *genotypes*, rather than the number of allele copies separating the individuals. For equally sized sampled real set R and synthetic set S, $d_{RR}(x)$ and $d_{RS}(x)$ denote the closest other-real and closest synthetic distances from $x \in R$, respectively, while $d_{SS}(y)$ and $d_{SR}(y)$ are the closest other-synthetic and closest real distances from $y \in S$. The implementation follows the authors' AATS equation (2023, p. 8):

$$AA_R = |R|^{-1}\sum_{x\in R}\mathbf{1}\{d_{RS}(x) > d_{RR}(x)\},\quad AA_S = |S|^{-1}\sum_{y\in S}\mathbf{1}\{d_{SR}(y) > d_{SS}(y)\},\quad AATS = \tfrac{1}{2}(AA_R+AA_S).$$

Ties contribute zero because the original equation uses a strict inequality. The nearest-neighbour chain experiment starts at every sampled row and repeatedly follows the nearest previously unseen row for four transitions, recording T/S sequences. As in the paper's Table 1, the number of real-versus-synthetic visits, not phase, determines the labels.

3.5 Privacy interpretation and missing inputs

If R_{train} contains verified training subjects and R_{test} contains subjects excluded from *all* model fitting, the paper reports the privacy-difference statistic $AATS(R_{\text{test}}, S) - AATS(R_{\text{train}}, S)$; its near-zero target is an empirical heuristic, not proof of privacy. The membership-attack command must be supplied with these two genotype CSVs plus one matching AFR/EUR/AMR condition vector per person and the saved generator/critic weights. For white-box ranking, $q(x)$ is the saved critic's score for candidate x conditioned on its ancestry vector; the adversary predicts membership for the highest u proportion of scores, with $u \in \{0.01, 0.10, 0.25, 0.50\}$. The reported precision is the fraction of true members among selected candidates. The black-box surrogate is intentionally described as a *surrogate* conditional PCA-WGAN critic trained on generated data and not necessarily an exact retraining of the original target critic, unlike the more specific setup in S15; see Hayes et al. [2019]. If holdout provenance or exact conditions are unavailable, these attacks are *not run* rather than manufacturing test labels.

4 Baseline WGAN-GP: what the objective means

A generator G_θ maps random latent noise $z \sim P_Z$ and ancestry condition c to a synthetic representation:

$$\tilde{x} = G_\theta(z, c).$$

The critic $C_\phi(x, c)$ returns an unrestricted real number rather than a probability. WGAN replaces a classical GAN classification objective with an integral probability metric related to Earth-mover/Wasserstein-1 distance [Arjovsky et al., 2017]. The conditional version supplies c to both networks, following the general conditional-GAN idea [Mirza and Osindero, 2014].

The critic is trained to make real samples score higher than generated samples. In the sign convention used by the code, the critic minimizes

$$L_C = \mathbb{E}[C_\phi(\tilde{x}, c)] - \mathbb{E}[C_\phi(x, c)] + \lambda_{\text{GP}} L_{\text{GP}},$$

and the generator minimizes

$$L_G = -\mathbb{E}[C_\phi(G_\theta(z, c), c)].$$

The gradient penalty is

$$L_{\text{GP}} = \mathbb{E}_{\hat{x}} \left(\|\nabla_{\hat{x}} C_\phi(\hat{x}, c)\|_2 - 1 \right)^2,$$

where $\hat{x} = \alpha x + (1 - \alpha)\tilde{x}$ and $\alpha \sim U(0, 1)$. This is the WGAN-GP construction of Gulrajani et al. [2017], used to encourage the critic to behave approximately as a 1-Lipschitz function along lines between real and generated samples.

5 Change 1: direct SNP-space generation instead of PCA reconstruction

5.1 What PCA does

Let centered real genotype vectors be rows of X_c. PCA diagonalizes the empirical covariance

$$\widehat{\Sigma} = \frac{1}{n-1} X_c^T X_c = U \Lambda U^T,$$

where the columns of U are eigenvectors (principal axes) and $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_p)$ contains descending eigenvalues. Population-genetic PCA is widely used to summarize ancestry/population structure [Patterson et al., 2006].

Retaining only the first k PCs produces the rank-k reconstruction

$$\widehat{X}_c = X_c U_k U_k^T.$$

The covariance represented by an idealized PCA reconstruction is

$$\widehat{\Sigma}_k = U_k \Lambda_k U_k^T.$$

Therefore all covariance carried exclusively by discarded components $k+1, \ldots, p$ is removed.

Proposition 1 (PCA truncation removes covariance in discarded eigendirections)**.** *Let $\Sigma = U \Lambda U^T$ and let $P_k = U_k U_k^T$. For a centered random genotype vector X with covariance Σ, the reconstruction $\widehat{X} = P_k X$ has covariance*

$$\mathrm{Cov}(\widehat{X}) = U_k \Lambda_k U_k^T.$$

Proof. Because P_k is deterministic and symmetric,

$$\mathrm{Cov}(P_k X) = P_k \,\mathrm{Cov}(X) P_k^T = P_k U \Lambda U^T P_k.$$

Using $P_k = U_k U_k^T$ and orthonormality $U_k^T U = [I_k \; 0]$ gives $U_k \Lambda_k U_k^T$. □

This does *not* imply that PCA is bad for population structure. It means that if local LD or other biologically important dependence lies partly in low-variance directions, a small k can discard it even when PC1–PC2 looks excellent.

5.2 What the independent binomial discretization does

The baseline obtains a continuous reconstructed dosage $V_j \in [0, 2]$ and then draws

$$G_j \mid V_j \sim \mathrm{Binomial}\left(2, \frac{V_j}{2}\right).$$

This is preferable to deterministic rounding for maintaining genotype variance, but it introduces locus-wise sampling noise.

Proposition 2 (Effect of independent binomial discretization on covariance)**.** *Assume that, conditional on the continuous vector V, genotype draws are independent across loci. For two distinct loci $j \neq k$,*

$$\mathrm{Cov}(G_j, G_k) = \mathrm{Cov}(V_j, V_k).$$

However,

$$\mathrm{Var}(G_j) = \mathrm{Var}(V_j) + \mathbb{E}\left[V_j\left(1 - \frac{V_j}{2}\right)\right].$$

Therefore the covariance can be retained in expectation while the marginal variances increase, which can attenuate correlation magnitudes.

Proof. By the law of total covariance,

$$\mathrm{Cov}(G_j, G_k) = \mathbb{E}[\mathrm{Cov}(G_j, G_k \mid V)] + \mathrm{Cov}(\mathbb{E}[G_j \mid V], \mathbb{E}[G_k \mid V]).$$

Conditional independence makes the first term zero. Since $\mathbb{E}[G_j \mid V] = V_j$, the second term is $\mathrm{Cov}(V_j, V_k)$. For the variance, the law of total variance gives

$$\mathrm{Var}(G_j) = \mathbb{E}[\mathrm{Var}(G_j \mid V)] + \mathrm{Var}(\mathbb{E}[G_j \mid V]),$$

and a Binomial$(2, V_j/2)$ variable has conditional variance $V_j(1 - V_j/2)$. □

The important conclusion is more nuanced than "independent sampling destroys all LD." It does not. Rather: (i) PCA truncation can already remove relevant covariance; (ii) independent discretization adds locus-specific noise and can reduce correlation coefficients; and (iii) no phase-specific haplotype pattern is explicitly modeled.

5.3 Why a 1-D convolution is a better inductive bias for ordered SNPs

A 1-D convolution with kernel width w computes local features

$$h_{t,r}^{(\ell+1)} = \sigma\left(b_r + \sum_s \sum_{u=0}^{w-1} K_{r,s,u} h_{t+u,s}^{(\ell)}\right).$$

Nearby SNPs contribute jointly to each hidden feature. Stacking convolutional layers increases the *receptive field*, the interval of input positions that can influence one hidden unit. For L stride-1 layers of width w, the receptive field is

$$R_L = 1 + L(w - 1).$$

Downsampling or dilation can expand it faster.

This architecture does not assume that adjacent SNPs are independent. It can learn motifs and local correlation patterns from ordered SNPs. Large-scale artificial-genome work has specifically used convolutional WGAN-GP architectures, residual blocks, positional information, and packing to better model large genomic regions [Yelmen et al., 2023]. Earlier work also evaluated artificial genomes by allele frequency, LD, haplotypic distances, and population structure [Yelmen et al., 2021].

5.4 Residual blocks and gradient flow

A residual block computes

$$y = x + F(x; \theta).$$

Its Jacobian is

$$\frac{\partial y}{\partial x} = I + J_F(x).$$

Thus backpropagated gradients have a direct identity route in addition to the learned transformation. This does not mathematically guarantee that gradients can never vanish, but it removes the requirement that every gradient path be a product only of learned Jacobians. Residual networks were introduced to make substantially deeper networks easier to optimize [He et al., 2016].

5.5 What the code implements

Use

```
--model_backend genotype_conv
--conv_channels 128
--conv_blocks 4
--pack_size 1
```

The direct model trains on imputed dosages $0, 1, 2$, maps them to $-1, 0, 1$, and lets the generator output continuous values in $[-1, 1]$. Generated continuous values are converted to expected dosage in $[0, 2]$ and then discretized by the same Binomial$(2, p)$ mechanism.

Important limitation: genotype is not haplotype. The current CSVs are unphased diploid dosages. A row such as $(1, 1)$ at two loci does not reveal whether the chromosomes are $(0, 0) + (1, 1)$ or $(0, 1) + (1, 0)$. Therefore no model trained only on dosage can recover true phase information that is absent from the input. The direct Conv1D backend can learn local *dosage* dependencies. Claims about haplotype fidelity require phased input and phase-aware evaluation. Yelmen et al.'s large-scale convolutional experiments used phased 0/1 haplotypes [Yelmen et al., 2023].

Memory scaling. A Conv1D activation tensor has approximate storage proportional to BCL, where B is batch size, C channel width, and L SNP count. For very large L, reduce `batch_size` or `conv_channels`. The implementation prints this warning explicitly.

6 Yelmen (2023) unconditional comparison backends: faithful networks and diploid adaptations

6.1 Scientific question and source-code provenance

The additional entry point `src/yelmen2023_pipeline.py` is a *separate, unconditional* benchmark against the existing PCA-space and SNP-space ancestry-conditioned WGANs. It uses the 2023 authors' 10K/65K WGAN generator and critic network definitions from the supplied `artificial_genomes-master/WGAN/models_10K.py` and `models_65K.py` and an optional 10K VAE from `VAE/models_10K_VAE.py` [Yelmen et al., 2023]. The port preserves the published 1-D hierarchical convolution, transposed-convolution, residual shortcuts, multiresolution Gaussian noise, trainable positional channels, and normalization layers. The training loop is packaged for reproducibility and correct formation of full packed critic minibatches. The categorical-visible RBM described below is *an explicit adaptation*, not a claim that the published CRBM already modeled unphased genotype dosage. The 2021 study demonstrated GAN/RBM population-structure and local-haplotype generation but did not provide reference-ancestry-conditioned models; it suggested conditional population control as future work [Yelmen et al., 2021].

6.2 Diploid input representation and information lost by unphasing

Let $N \geq 2$ denote the number of individuals selected for training, $M \geq 1$ denote the number of ordered biallelic SNP sites, $i \in \{1, \ldots, N\}$ index individuals, and $j \in \{1, \ldots, M\}$ index sites. Let $G_{ij} \in \{0, 1, 2\}$ count the number of copies of the ALT allele at site j carried by individual i; this is the *unphased additive genotype dosage*. The direct convolutional networks use $X_{ij} = G_{ij}/2 \in \{0, 1/2, 1\}$. For model comparison, a missing dosage (encoded -1 in the audited CSVs) is filled by the mean

dosage at that SNP among selected training individuals. If every selected training individual is missing at SNP j, the existing cWGAN uses a zero fill and the new Yelmen default reproduces that convention. The cWGAN estimates query and reference means separately after selecting their shared SNPs, whereas the unconditional benchmarks do not use reference genotypes. SNPs are ordered by genomic position in the Yelmen implementation; the cWGAN retains the reference-file column order. Both decisions are documented in QC files. The original phased input instead had one binary allele per row and locus. For a diploid double heterozygote, the $(0,0)/(1,1)$ and $(0,1)/(1,0)$ haplotype configurations yield identical dosages and cannot be distinguished from G; the new models consequently learn *genotype* dependencies, not hidden phase. Let p_j denote the ALT-allele frequency at SNP j, estimated in training by $\widehat{p}_j = (2N)^{-1} \sum_{i=1}^{N} G_{ij}$, with reference allele frequency $1 - \widehat{p}_j$.

6.3 Input auditing and matched imputation for the model comparison

Let M_{ij} be the missing-call indicator for individual i at SNP j, equal to 1 if the genotype is missing and 0 otherwise. The audited CSVs encode missing calls as -1, while observed hard calls are $G_{ij} \in \{0,1,2\}$. Define $O_j = \{i \in \mathcal{T} : M_{ij} = 0\}$, where $\mathcal{T}$ is the index set of individuals selected for training and $|O_j|$ is the number of called genotypes at SNP j among those individuals. The per-SNP mean and imputed dosage are

$$\widehat{\mu}_j = \begin{cases} |O_j|^{-1} \sum_{i \in O_j} G_{ij}, & |O_j| > 0, \\ 0, & |O_j| = 0, \end{cases} \qquad \widetilde{G}_{ij} = \begin{cases} G_{ij}, & M_{ij} = 0, \\ \widehat{\mu}_j, & M_{ij} = 1. \end{cases}$$

The zero fallback for $|O_j| = 0$ is an implementation convention, *not* evidence that the true genotype is homozygous reference. It must be reported because it can induce large biases in the very sparse SAI and VDC3 cohorts. Held-out individuals, if present, receive $\widehat{\mu}_j$ estimated from training individuals only. The cWGAN applies the same averaging rule separately to its query and ancestry-reference matrices after intersecting SNP columns; the new Yelmen default uses the query cohort only. The Yelmen subset-selection permutation is now matched to the cWGAN by hashing the seed and dataset identifier, so n-sample sweeps select the same individuals when original CSV rows and cohort identifiers agree. Nevertheless, the SNP column sets need not match automatically: the cWGAN restricts sites to those shared with its reference panel, while the Yelmen code trains on the query CSV SNPs, sorted by chromosome. For a controlled model comparison, the training SNP intersection must be harmonized before fitting. For the original cWGAN, only negative values trigger imputation; NaN values remain unhandled and are flagged by the audit. The Yelmen loader converts recognized missing sentinels and blanks to NaN before mean imputation; for the audited $-1/0/1/2$ data, the two rules agree. An optional modal-imputation policy is retained as an ablation, and finite fractional dosages are preserved by default.

The categorical RBM cannot ingest arbitrary $\widetilde{G}_{ij} \in [0,2]$ because its visible state space has three categories. For that model only, define the hard-call mapping $C_{ij} = \min(2, \max(0, \lfloor \widetilde{G}_{ij} + \frac{1}{2} \rfloor)) \in \{0,1,2\}$. This changes imputed calls (and any originally fractional dosages) and is reported separately from the WGAN/VAE input. Thus *matched mean imputation* does not imply identical representation in all three networks. The two training CLIs print counts of genotype 0, 1, 2, missing -1, other sentinels, and fractional calls and save stagewise JSON, CSV and per-SNP QC records in `input_qc/`. They warn for missingness above 10% and absent genotype-0 calls; `--max_missing_fraction` and `--fail_on_absent_zero` enable optional rejection without changing default training behavior. The former is an ad hoc engineering threshold and not a population-genetic significance criterion. No SNP is discarded automatically by these new checks. As in the supplied Yelmen source and the

retained cWGAN, mean or modal imputation is only an input approximation; neither should be interpreted as recovery of unobserved alleles. For the integer-only genotype-motif and heterozygosity evaluations, a separate evaluation matrix is constructed from each mean-imputed real matrix by replacing non-integer entries with the SNPwise modal observed hard call, with a zero fallback if no hard call exists. This mirrors the previously delivered cWGAN retrospective evaluator and is now shared by both model comparisons; the conversion is recorded and does not alter model-training inputs. Any comparison should use this common integer-evaluation protocol rather than mix dosage-based and category-based real-data references.

6.4 Original multiscale convolutional WGAN-GP with minimal dosage changes

Let z denote the initial Gaussian noise tensor and u_ℓ denote an independent Gaussian tensor injected into upsampling scale ℓ; the authors' generator $F_\theta(z, u_1, \ldots, u_K)$ returns a sigmoid output $Y_j \in (0, 1)$ at each locus. Here θ denotes generator trainable parameters and K denotes the number of additional noise resolutions, as fixed by the supplied 10K/65K source. The *unphased* output genotype is $\widetilde{G}_j = \min\{2, \max\{0, \lfloor 2Y_j + 1/2 \rfloor\}\}$; this is a post-training nearest-integer discretization, not a binomial trial and not a trained categorical head. The code can optionally apply the existing five genotype discretizers without changing the generator. The discriminator (critic) C_ϕ, with trainable parameters ϕ, processes a *pack* of $m = 3$ independent genotype vectors as one item. The training criterion is

$$L_C(\phi) = \mathbb{E}[C_\phi(\widetilde{\mathbf{X}})] - \mathbb{E}[C_\phi(\mathbf{X})] + \lambda \mathbb{E}_{\epsilon, \mathbf{X}, \widetilde{\mathbf{X}}} \left[\left(\|\nabla_{\widehat{\mathbf{X}}} C_\phi(\widehat{\mathbf{X}})\|_2 - 1 \right)^2 \right], \tag{1}$$

$$L_F(\theta) = -\mathbb{E}[C_\phi(\widetilde{\mathbf{X}})], \qquad \widehat{\mathbf{X}} = \epsilon \mathbf{X} + (1 - \epsilon)\widetilde{\mathbf{X}}, \quad \epsilon \sim \text{Uniform}(0, 1). \tag{2}$$

Here $\mathbf{X}$ is a randomly assembled pack of m real normalized genotype vectors, $\widetilde{\mathbf{X}}$ an equally sized pack of generated vectors, ϵ an independent scalar drawn uniformly on $[0, 1]$ for each pack, $\widehat{\mathbf{X}}$ the point interpolated between these packs, ∇ the gradient with respect to the interpolated pack, $\|\cdot\|_2$ Euclidean norm, $\mathbb{E}$ expectation, and $\lambda = 10$ the gradient-penalty coefficient. The critic minimizes L_C and the generator minimizes L_F, with ten critic updates per generator update. The supplied training script also applies an optional random $[0, 0.1]$ perturbation to the real-score multiplier; the new flag `--wgan_real_label_noise` reproduces this source-level detail and `--no-wgan_real_label_noise` disables it, yielding exactly the stated unperturbed objective. The optional second-stage fine-tuning flag `--wgan_finetune_epochs` continues the same WGAN weights with a default lower generator learning rate of 5×10^{-5}, matching the reduced-rate refinement described for the 65K study; the implementation reinitializes the Adam optimizer states at this stage. The Adam learning rates are 0.0005 for each network and its momentum parameters are $(\beta_1, \beta_2) = (0.5, 0.9)$; the WGAN defaults to 96 packs per minibatch in the source command configuration (`WGAN/main_WGAN.py`). Unlike the existing `genotype_conv` model, this architecture has no ancestry input or FiLM layer, uses multiple latent-noise resolutions, has trainable positional channels throughout, and matches the source's convolution/transposed-convolution hierarchy. Network lengths are kept fixed at 16,383 or 65,535 sites. Shorter segments are right-padded with zero-valued pseudo-sites; the generated pseudo-sites are removed from the CSV. Longer chromosomes are split into independently trained windows; this limitation must be reported in any chromosome-wide inference.

6.5 Three-state visible RBM and conditional overlap

The publication's RBM has one *binary allele* visible unit per position. For an unphased genotype, let $g = (g_1, \ldots, g_V)$ denote one genomic window of V SNPs with $g_j \in \{0, 1, 2\}$. Let $h = (h_1, \ldots, h_H)$

denote H binary hidden units, $h_a \in \{0,1\}$; $H = 1000$ by the large-CRBM paper preset. Define $a_{jk} \in \mathbb{R}$ as the visible bias for genotype category $k \in \{0,1,2\}$ at SNP j, $b_a \in \mathbb{R}$ as hidden-unit a's bias, and $W_{jka} \in \mathbb{R}$ as the coupling between that category and hidden unit. The categorical–Bernoulli RBM has the energy

$$E(g,h) = -\sum_{j=1}^{V} a_{j,g_j} - \sum_{a=1}^{H} b_a h_a - \sum_{j=1}^{V}\sum_{a=1}^{H} W_{j,g_j,a} h_a, \qquad p(g,h) = \frac{e^{-E(g,h)}}{Z},$$

where $Z = \sum_{g\in\{0,1,2\}^V} \sum_{h\in\{0,1\}^H} e^{-E(g,h)}$ is the finite partition function. Its Gibbs conditional probabilities are

$$\Pr(g_j = k \mid h) = \frac{\exp(a_{jk} + \sum_a W_{jka} h_a)}{\sum_{r=0}^{2} \exp(a_{jr} + \sum_a W_{jra} h_a)}, \qquad \Pr(h_a = 1 \mid g) = \sigma\left(b_a + \sum_j W_{j,g_j,a}\right),$$

where $\sigma(v) = 1/(1+e^{-v})$ is the sigmoid. These equations replace the original Bernoulli visible-unit conditional by a three-way softmax; the hidden conditional and bipartite energy-based structure remain unchanged.

For a conditional window, partition its visible-site index set into P (pinned observed context) and I (to be inferred), with $P \cap I = \varnothing$ and $P \cup I = \{1,\ldots,V\}$. The CRBM maximizes $\log p(g_I \mid g_P)$, using a positive conditional expectation over hidden units and a negative expectation over Gibbs chains that keep g_P fixed. *Out-of-equilibrium* (OOE) training means that every negative chain starts afresh from a fixed initial distribution $p_0(g_I)$ and runs a fixed number k of alternating hidden/visible Gibbs updates, rather than initializing at a training observation (CD-k) or carrying a persistent chain across updates (PCD). Here p_0 assigns probability 1/3 to each genotype category at each free visible unit, instead of the publication's Bernoulli(1/2) initial allele distribution. The large-data defaults are $k = 100$, learning rate 0.005, and $H = 1000$ [Yelmen et al., 2023]. No gradient is applied to the biases of the pinned visible variables because those biases cancel in $p(g_I \mid g_P)$; pinned-to-hidden couplings remain trainable. The first block generates up to 5,000 SNP genotypes. Each following block conditions on up to 5,000 previously generated SNPs and infers the next at most 5,000, matching the published overlap principle. The new categorical visible parameterization has approximately $3VH$ interaction weights per window, versus VH in the binary-allele RBM, and can therefore use much more memory.

6.6 Optional VAE benchmark and non-equivalence of likelihoods

The optional convolutional VAE uses the authors' 16,383-position encoder/decoder, retaining the Gaussian reparameterization and KL regularization but removing a hard-coded CUDA allocation in the supplied source. Let $q_\psi(z \mid X)$ denote the encoder's approximate Gaussian posterior with parameters ψ, let $p(z) = \mathcal{N}(0, I)$ be the standard normal prior, and let $\widehat{X}$ denote the sigmoid decoder's predicted normalized genotype. The implemented objective is a summed binary cross-entropy between $X \in [0,1]$ and $\widehat{X}$, plus $\mathrm{KL}(q_\psi(z \mid X) \| p(z))$, where KL denotes Kullback–Leibler divergence. It is deliberately close to the source VAE but is not a proper learned three-category genotype likelihood. This benchmark is optional because the published large-SNP VAE could not be made satisfactory [Yelmen et al., 2023].

6.7 Comparison design, selected checkpoints and run organization

The script uses the existing multi-cohort `--datasets`, `--dataset_inputNAME=PATH`, `--subset_samples`, and `--seed` conventions but accepts *no* `--ref_input`, `--fam`, `--condition_source`,

or `--target_admixture`. Each dataset/subset/model obtains separate checkpoints and genotype CSVs under `runs/<timestamp>_<label>/<cohort>_<subset>/yelmen_<model>/`. A consolidated `checkpoint_baseline.pt`, `checkpoint_best.pt`, or `checkpoint_last.pt` contains the state of *every* independently trained genomic segment. By default, `--save_modelsbaselinebestlast` saves an *untrained* `baseline` at epoch 0, captured after initialization and before any critic, generator, or RBM update. WGAN baseline bundles include both generator and critic parameters; CRBM baseline bundles store the initial state of every genomic window, including the pinned-context model. The optional `first` means the state after epoch 1 and is not interchangeable with `baseline`. Baseline-generated data are stored in `saved_model_outputs/epoch_00000_baseline/`. For comparability, the SNP column names are retained but reordered to chromosome and increasing base-pair position, and the zero-based training row indices are archived. The automated `best` criterion is the sum of two *in-sample* diagnostics: mean absolute ALT-frequency error across generated versus observed SNPs and absolute mean heterozygote-frequency error. Thus `best` is reproducible but is *not* the original authors' visually chosen PCA stopping epoch and *not* out-of-sample validation. An explicit holdout fraction excludes individuals from training so their distance/privacy diagnostic can be computed separately; no differential-privacy guarantee follows from a favorable score.

The existing `genotype_paper_evaluation.py` supplies the same unphased-dose adaptations for allele-frequency fidelity, observed and HWE-expected heterozygosity, PCA, physical-distance LD by chromosome and pooled within-chromosome LD, centered three-point dosage moments, three-state SNP motifs, AATS, and nearest-neighbor chains. In `--evaluate_run` mode, the benchmark reads `synthetic_genotypes.csv` by default *without opening its model checkpoint*; `--resample` explicitly reloads requested weights and writes additional method- and seed-labeled outputs, preserving the canonical CSV. For individually trained chromosomes or nonoverlapping WGAN windows, concatenation assembles independent blocks but does *not* preserve dependence across their boundaries, including genome-wide ancestry correlations. The experimental comparison must report the advantage the existing cWGAN gets from its reference panel and ancestry conditions, the very small cohort sizes in sample-size sweeps, and differences between dense haplotypes in the 2023 study and LD-pruned genotype SNPs in this project.

```
python src/yelmen2023_pipeline.py --datasets BOG \
  --dataset_input BOG=data/BOG/unphased.csv \
  --models wgan crbm --genome_scope per_chromosome \
  --subset_samples "{'BOG':[25,100,'all']}" \
  --save_models baseline best last --seed 42
# Default: analyze existing saved genotype CSVs
python src/yelmen2023_pipeline.py --evaluate_run runs/RUN_NAME \
  --real_input BOG=data/BOG/unphased.csv
# Explicit resampling (new files; canonical output unchanged)
python src/yelmen2023_pipeline.py --evaluate_run runs/RUN_NAME \
  --real_input BOG=data/BOG/unphased.csv --resample \
  --weights best --sampling_methods nearest categorical
```

7 Checkpoint-compatible discrete-genotype sampling experiments

7.1 Why clipping deserves a controlled comparison

Let $Y_{ij} \in \mathbb{R}$ be the generator's continuous post-inverse-PCA dosage prediction for generated individual i at SNP j (or the restored direct-convolution output). For a PCA backend, the original procedure maps it to $P_{ij} = \min(1, \max(0, Y_{ij}/2))$ and draws the observed genotype $G_{ij} \sim$

Binomial$(2, P_{ij})$. Here the binomial variable counts two independent Bernoulli trials with the same success probability P_{ij}. This construction preserves expected dosage $2P_{ij}$, but clipping makes $Y_{ij} = 3$ and $Y_{ij} = 4$ both map to the same $P_{ij} = 1$, and independent draws cannot reproduce a general genotype probability vector. Within a population with heterozygote excess or deficit, the conditional genotype distribution need not equal a binomial distribution. However, a binomial draw is *not* inherently a phase-specific model: it can be a valid unphased genotype model under HWE. The code keeps this baseline untouched so its scientific assumptions remain testable.

7.2 Five post-hoc samplers and their exact definitions

Let $\tau > 0$ be a user-specified softness/temperature and y be a single raw Y_{ij} value. The implementations are:

(i) `binomial`: $G \sim \text{Binomial}(2, \min(1, \max(0, y/2)))$ (historical reference).

(ii) `nearest`: $G = \min(2, \max(0, \lfloor y + \frac{1}{2} \rfloor))$ (deterministic nearest integer with half-up ties).

(iii) `categorical`: for $g \in \{0, 1, 2\}$ let $\pi_g(y) = \exp[-(y-g)^2/(2\tau^2)] / \sum_{h=0}^{2} \exp[-(y-h)^2/(2\tau^2)]$ and sample G from the categorical distribution $\Pr(G = g \mid y) = \pi_g(y)$. This is a post-hoc radial-basis softmax, *not* a learned three-logit generator head.

(iv) `ternary`: set $U = \tanh[(y-1)/\tau] \in (-1, 1)$; report $G = 0$ when $U < -1/2$, $G = 1$ when $-1/2 \le U \le 1/2$, and $G = 2$ when $U > 1/2$. This rounds in the centred interval $[-1, 1]$ using fixed thresholds.

(v) `ternary_normal`: first add independent noise $\varepsilon \sim N(0, \tau^2)$ to U, clip $U + \varepsilon$ into $[-1, 1]$, then use those same thresholds. $N(0, \tau^2)$ denotes a Gaussian with mean zero and variance τ^2.

Only the binomial, categorical and noisy-ternary methods are stochastic. These methods are comparators, not claims of superior biological fidelity, and all preserve compatibility with earlier weights. A genuine *learned* categorical head would require architectural change, retraining and a differentiable categorical training objective; saved scalar-output checkpoints do not contain its three logits. The raw histogram, per-SNP table, and JSON quantify the proportions with $Y < 0$ and $Y > 2$ and report extrema, quantiles and absolute clipping effects *before* division, clipping, or categorization. Generated samples from a shared checkpoint use the same conditions and latent draw seed across sampler methods.

7.3 Run routing and output semantics

With no `--evaluate_run`, the main command trains and samples saved `baseline`, `best` and/or `last` weights. `--no_evaluate` opts out of paper-style analysis; evaluation is enabled by default. Each selected state writes conventional `outputs/synthetic_genotypes.csv`, optional `synthetic_genotypes_METHOD.csv`, and `sampling_diagnostics/preclip_dosage_distribution.png`. The canonical CSV is generated by the *first* method in `--sampling_methods`; this is recorded in CLI arguments. Each analysis is placed in `paper_evaluation/METHOD/plots/` and `paper_evaluation/METHOD/metrics/`. Retrospective `--evaluate_run` discovers existing saved CSVs without retraining. Its optional `-resample` loads the requested checkpoint, its PCA basis and archived ancestry conditions; it creates additional method-specific CSVs and diagnostic plots. Older saved PCA files may lack the post-sign-flip provenance marker; these are warned about and should be validated against the original synthetic CSV before interpreting resampling as bit-for-bit reproduction.

7.4 Sampler-aware checkpoint selection and unclipped PCA/dosage comparisons (October 2026)

The baseline cWGAN uses *continuous* PC coordinates or dosages in the WGAN–GP training objective; discrete genotype sampling is a downstream operation. For model selection, however, the genotype representation is consequential. At evaluated epoch t, the same seeded latent and ancestry draws yield a continuous synthetic dosage matrix $Y^{(t)}$. Given a chosen sampler s and its temperature τ, compute

$$\widetilde{G}^{(t,s)} = S_s(Y^{(t)}; \tau, \omega), \qquad \widehat{t}_s = \arg\min_{t \in \mathcal{T}} Q(\widetilde{G}^{(t,s)}, G_{\text{comparison}}),$$

where Q is the existing composite or PC-MMD^2 checkpoint score, $\mathcal{T}$ contains the evaluated epochs, and ω denotes a fixed random seed reused at every scored epoch. The CLI parameter `--checkpoint_sampling_method` determines s; when omitted, s is the *first* entry in `--sampling_methods`, binomial by default. This changes the selection of `best`, not the gradient computation or the original WGAN architecture. The selection record logs s and τ. A method-specific best epoch need not equal that found using binomial sampling; choosing it requires training-time evaluations or preserved per-epoch checkpoints. Sampling from an already saved `best` checkpoint cannot reconstruct a different optimum ex post.

To distinguish range violations from stochastic genotype noise, define

$$C_{ij} = \min\{2, \max\{0, Y_{ij}\}\}, \quad f_{\text{clip}} = \frac{1}{nm} \sum_{i,j} \mathbf{1}\{Y_{ij} < 0 \text{ or } Y_{ij} > 2\}, \quad \delta_{\text{clip}} = \frac{1}{nm} \sum_{i,j} |Y_{ij} - C_{ij}|.$$

The script saves the preclipping histogram, histogram of C, summary quantiles, f_{clip}, δ_{clip}, and the fraction of altered values at each SNP. This is distinct from sampling variance, which can remain substantial even when $f_{\text{clip}} = 0$.

Let $P(\cdot)$ denote the PCA projection fitted on the reference panel. For each sampler, compare the actual discrete synthetic genotypes $P(\widetilde{G}^{(s)})$ with both the continuous projected values $P(Y)$ and the real target subjects projected in the *same* PCA basis. Method-specific PC1–PC2 overlays, PC1–PC6 histograms, and per-PC mean and standard-deviation shifts are saved. Unlike the historical direct-genotype convolutional `pca_overlay.png`, the new canonical overlay is computed from the actual sampled 0/1/2 CSV. The continuous projection is retained as `pca_overlay_continuous.png`.

During new training, these outputs belong to each existing `saved_model_outputs/epoch_XXXXX_LABELS/` folder. For archived cWGAN weights, explicit `--evaluate_run...--resample` reuses the archived PCA model and ancestry conditions, writes a new non-overwriting `sampling_experiments/NAME/` directory under each selected checkpoint, and records sampler, seed, weight provenance and real-data imputation. `--plots_only` skips expensive paper-style analyses. All sampling-method panels use shared latent and ancestry draws, so their differences are attributable to post-processing for fixed model parameters. Reconstruction from legacy PCA checkpoints must still account for the recorded PC orientation, and the original query subjects/variant order must remain available. Existing saved weights cannot be reselected as method-specific historical `best` without retraining or a complete archive of evaluated epoch states.

8 Change 2: repeated ancestry conditioning with FiLM

8.1 Why input-only concatenation can be weak

The baseline concatenates condition c to the generator's latent vector at the input, $[z, c]$, and to the critic input, $[x, c]$. This is a valid conditional architecture and, with enough capacity, can approximate

rich conditional functions. The problem is not representational impossibility. The practical concern is that information from c must propagate through every subsequent transformation.

For a depth-L network $h_L = f_L \circ \cdots \circ f_1([z, c])$, the derivative of a late hidden representation with respect to c contains a product of Jacobians:

$$\frac{\partial h_L}{\partial c} = J_{f_L} J_{f_{L-1}} \cdots J_{f_1, c}.$$

If several factors contract directions associated with c, sensitivity to the condition can become small.

8.2 FiLM definition

Feature-wise Linear Modulation (FiLM) applies an affine transformation to hidden features whose parameters are functions of the condition [Perez et al., 2018]:

$$\mathrm{FiLM}(h; c) = \gamma(c) \odot h + \beta(c).$$

For the implemented MLP, both $\gamma(c)$ and $\beta(c)$ are learned affine maps. The layer is initialized near identity using $\gamma(c) = 1 + \Delta\gamma(c)$ with $\Delta\gamma = 0$ and $\beta = 0$ initially.

With FiLM inserted at multiple depths, the final output has multiple direct computational paths from c. Schematically,

$$\frac{\partial h_L}{\partial c} = \text{long path from early layers} + \sum_{\ell=1}^{L} \text{path from FiLM injection at layer } \ell.$$

This does not prove FiLM must outperform concatenation. It gives the condition repeated opportunities to modulate features rather than relying on a single input injection.

8.3 Code

Use

```
--model_backend pca_film
```

This leaves the PCA representation unchanged and is therefore the cleanest first architecture ablation.

9 Change 3: external ancestry coefficients instead of PCA-barycentric conditions

9.1 The current barycentric construction

Let $v_{AFR}, v_{EUR}, v_{AMR} \in \mathbb{R}^k$ be reference-population centroids in PC space. The current condition for a query sample with PC coordinate y approximately solves

$$y \approx q_{AFR} v_{AFR} + q_{EUR} v_{EUR} + q_{AMR} v_{AMR}, \qquad \sum_k q_k = 1,$$

by least squares, followed by clipping negative coefficients and renormalization.

This is geometrically intuitive, but it makes the condition a deterministic function of the same PC representation that the baseline generator is trained to generate. Consequently, success at generating PC coordinates consistent with c is partly built into how c was defined. This is not "data leakage" in the machine-learning train/test sense; it is a representational circularity that can make conditional interpretation less independent.

9.2 ADMIXTURE-style model-based ancestry

ADMIXTURE estimates global ancestry using a generative allele-frequency model rather than defining ancestry as coordinates inside the PCA space [Alexander et al., 2009]. A simplified formulation is

$$X_{ij} \sim \text{Binomial}(2, \pi_{ij}), \qquad \pi_{ij} = \sum_{k=1}^{K} q_{ik} f_{kj},$$

where q_{ik} is individual i's ancestry fraction from source k and f_{kj} is the alternative-allele frequency of SNP j in source k.

PCA and ADMIXTURE answer related but different questions: PCA provides axes summarizing covariance/population structure, while model-based ancestry methods directly estimate mixture coefficients under a specified latent-population model. Alexander et al. explicitly contrast PCA-based approaches with model-based ancestry estimation [Alexander et al., 2009].

Important nuance. If ADMIXTURE and PCA are computed from the same genotype matrix, the estimates are not statistically independent data. The advantage is that the condition is no longer defined as a direct barycentric transformation of the model's output coordinates.

9.3 Code

The pipeline accepts either CSV/TSV files with named columns `AFR,EUR,AMR` or a headerless three-column ADMIXTURE-like `.Q` file:

```
--condition_source external \
--condition_input BOG=data/BOG/BOG.Q \
--external_condition_order AFR,EUR,AMR
```

For combined models, pass one `--condition_inputNAME=PATH` per constituent dataset. Row order must exactly match the genotype CSV row order. The loader checks row counts, non-negativity, finite values, and normalizes every row to the simplex.

10 Change 4: population-balanced minibatch sampling

10.1 The imbalance problem

Suppose the training set contains groups $g = 1, \ldots, G$ with sizes n_g. Uniformly sampling individuals optimizes an empirical objective of the form

$$L_{\text{emp}}(\theta) = \sum_{g=1}^{G} \frac{n_g}{N} \left(\frac{1}{n_g} \sum_{i \in g} \ell_i(\theta) \right), \qquad N = \sum_{g} n_g.$$

Large groups receive larger coefficients n_g/N. This is appropriate if the desired training distribution is exactly the observed pooled sample mixture. It is not appropriate if the goal is to give a small admixed query cohort comparable influence to a much larger reference set.

10.2 Inverse-frequency weighted sampling

Assign individual i in group $g(i)$ sampling weight

$$w_i \propto \frac{1}{n_{g(i)}}.$$

Then the total sampling mass of each group is

$$\sum_{i \in g} w_i \propto n_g \frac{1}{n_g} = 1.$$

After normalization, each group has probability $1/G$. Therefore the expected stochastic gradient corresponds to the group-balanced objective

$$L_{\text{bal}}(\theta) = \frac{1}{G} \sum_{g=1}^{G} \left(\frac{1}{n_g} \sum_{i \in g} \ell_i(\theta) \right).$$

This is the precise sense in which the sampler "balances" the data. It changes the optimization target; it does not create additional biological information.

10.3 Code and groups

Use

```
--balanced_sampling
```

with `pca_film` or `genotype_conv`. Query groups are labeled separately (for example `QUERY_BOG`, `QUERY_BAQ`); reference groups are `REF_AFR`, `REF_EUR`, `REF_AMR`, and `REF_OTHER` if present. The run log prints observed group counts.

When not to use it. If the scientific target is the empirical pooled population distribution, equalizing groups biases the training objective away from that distribution. Treat balanced sampling as a hypothesis to test, not an automatic correction.

11 Change 5: PacGAN-style critic packing to probe mode collapse

11.1 Mode collapse

Let P be the real distribution and Q_θ the generator distribution. *Mode collapse* means Q_θ places too little or zero probability on regions that have appreciable probability under P, even if generated samples individually look plausible.

11.2 Packing

PacGAN changes the critic from a function of one sample to a function of a pack of m samples [Lin et al., 2018]:

$$C(x_1, \ldots, x_m, c_1, \ldots, c_m).$$

The real and generated distributions seen by the critic become product distributions P^m and Q^m.

A simple calculation shows why missing modes can become easier to detect. Suppose a region A has real mass $P(A) = \varepsilon > 0$ but the collapsed generator has $Q(A) = 0$. In one sample, the probability of seeing the missing region under P is ε. In a pack of m independent real samples, the probability that at least one lies in A is

$$1 - (1 - \varepsilon)^m,$$

whereas the generator's probability remains 0. Since $1 - (1 - \varepsilon)^m$ increases with m, the missing mode creates a stronger pack-level signal.

11.3 Code

Use, for example,

```
--pack_size 2
```

with `pca_film` or `genotype_conv`. In vector space, packed samples are concatenated along the feature dimension. In Conv1D space, packed samples become input channels to the critic. Conditions are packed consistently with their samples. The gradient penalty is then computed with respect to the packed critic input.

Packing reduces the number of independent critic decisions per minibatch by a factor of m. Very large m can therefore increase variance or memory use; $m = 2$ is a reasonable first ablation.

12 Change 6: multi-metric model selection instead of selecting by GAN loss alone

12.1 Why adversarial loss is insufficient

WGAN losses are useful training diagnostics [Arjovsky et al., 2017], but a low adversarial objective does not logically imply correct MAF, LD, heterozygosity, population structure, or privacy. Genomic synthetic-data studies evaluate multiple properties because no single statistic captures all biologically relevant structure [Yelmen et al., 2021, 2023, Wharrie et al., 2023].

The code already records MAF error, heterozygosity, dosage-LD error, RBF-MMD in PC space, and nearest-neighbor adversarial accuracy. MMD is a distributional two-sample discrepancy in a reproducing-kernel Hilbert space [Gretton et al., 2012].

12.2 Loss orientation

The selector converts all metrics to "smaller is better" losses:

$$\begin{aligned}
\ell_{MAF} &= \mathrm{MAE}(MAF_{real}, MAF_{syn}),\\
\ell_{H} &= |\overline{H}_{real} - \overline{H}_{syn}|,\\
\ell_{LD} &= \mathrm{MAE}(r^2_{real}, r^2_{syn}),\\
\ell_{MMD} &= \mathrm{MMD}^2,\\
\ell_{AA} &= |AA - 0.5|.
\end{aligned}$$

For the nearest-neighbor adversarial-accuracy statistic, 0.5 corresponds to chance-level local distinguishability, hence the absolute distance from 0.5.

12.3 Why scaling is necessary

These metrics have different numerical units and ranges. Adding raw values would make the largest-scale metric dominate arbitrarily. Across candidate models, the selection script estimates a robust scale s_j for each metric using its interquartile range (falling back to median absolute deviation when needed), then computes

$$S_i = \frac{\sum_j w_j \ell_{ij}/s_j}{\sum_j w_j}.$$

Lower S_i is better under the chosen weights.

This score is *candidate-set dependent*; it is not a universal biological-quality number. It should compare models evaluated on the same SNPs and the same independent holdout cohort.

12.4 Code

```
python select_best_cwgan_run.py 'runs/20261003_*' \
    --output model_comparison.csv
```

Weights can be overridden with repeated `-weight METRIC=VALUE`. The shared trainer in `cwgan_extensions/models.py` also exposes an `epoch_callback` interface that can restore the best epoch under an externally supplied selection score. It is deliberately not enabled by default because a checkpoint score requires a fixed validation protocol and explicit metric scaling choices.

13 Change 7: privacy evaluation separated from fidelity evaluation

13.1 Fidelity and privacy are different questions

A model that exactly memorizes the training set can have excellent MAF, LD, and PCA fidelity. Therefore fidelity cannot prove privacy. Recent genomic synthetic-data evaluations include relatedness, nearest-neighbor, membership-inference, and multi-representation leakage analyses [Yelmen et al., 2023, Wharrie et al., 2023, Correa Rojo et al., 2026].

13.2 Exact replay

An exact replay occurs if a synthetic genotype row is byte-for-byte identical to a real genotype row. The code reports the fraction of training and holdout real rows that appear exactly in the synthetic set. Zero exact replay is reassuring but extremely weak: a nearly copied genome is not an exact replay.

13.3 Distance-based membership inference

Let $\phi(x)$ be a standardized PCA embedding and let S be the synthetic set. Define

$$d(x, S) = \min_{y \in S} \|\phi(x) - \phi(y)\|_2^2, \qquad s(x) = -d(x, S).$$

If training members are systematically closer to the synthetic set than real non-members held out from training, $s(x)$ can reveal membership. The script reports ROC AUC for distinguishing training members from holdout non-members. AUC $= 0.5$ means this particular attack is at chance; it does not establish privacy.

13.4 Differential privacy is a stronger formal guarantee

A randomized training algorithm $\mathcal{A}$ is (ε, δ)-differentially private if, for every pair of neighboring datasets D, D' differing in one individual and every measurable output set S,

$$\Pr[\mathcal{A}(D) \in S] \leq e^{\varepsilon} \Pr[\mathcal{A}(D') \in S] + \delta.$$

None of the diagnostics implemented here proves this inequality. They are empirical audits for memorization/proximity leakage only.

13.5 Code

Privacy evaluation requires a real training set and a genuinely unused real holdout set:

```
python evaluate_genotype_privacy.py \
  --train_real real_train.csv \
  --holdout_real real_holdout.csv \
  --synthetic runs/.../outputs/synthetic_genotypes.csv \
  --output privacy_metrics.json
```

Do not use the same individuals in `train_real` and `holdout_real`.

14 Relationship between the changes

The changes target different failure modes and should not be treated as substitutes.

Change	Failure mode targeted	What improvement would look like
Direct Conv1D SNP model	PCA rank truncation; weak explicit local-locus inductive bias	Better LD/MAF and higher-order local dependence without loss of ancestry structure
FiLM conditioning	Condition influence may weaken through depth	Better ancestry-conditional fidelity at matched unconditional quality
External ancestry q	Condition defined directly from output PC geometry	Conditional behavior remains good under independently modeled ancestry coefficients
Balanced sampling	Large groups dominate SGD expectation	Better target-query fidelity without unacceptable loss on reference structure
Packing	Generator covers too few modes	Better diversity/coverage and adversarial accuracy without frequency/LD degradation
Multi-metric selection	GAN loss does not encode biological fidelity	Selected runs improve several holdout metrics simultaneously
Privacy audit	High-fidelity memorization can go unnoticed	Low replay/proximity/membership signal while retaining utility

15 Concrete command-line experiments

15.1 Baseline reproduction

```
python conditional_wgan_pipeline.py \
  --datasets BOG \
  --dataset_input BOG=data/BOG/genotypes.csv \
  --ref_input data/reference/reference.csv \
  --fam data/reference/reference.fam \
  --model_backend baseline_pca \
  --val_fraction 0
```

Setting `val_fraction=0` restores use of all samples for training and disables a real validation split. The original code did not use training data "as validation"; it simply had no validation set.

15.2 FiLM only

```
python conditional_wgan_pipeline.py ... \
  --model_backend pca_film \
  --pack_size 1
```

15.3 FiLM plus balanced sampling

```
python conditional_wgan_pipeline.py ... \
  --model_backend pca_film \
  --balanced_sampling
```

15.4 FiLM plus PacGAN packing

```
python conditional_wgan_pipeline.py ... \
  --model_backend pca_film \
  --pack_size 2
```

15.5 External ancestry coefficients

```
python conditional_wgan_pipeline.py ... \
  --model_backend pca_film \
  --condition_source external \
  --condition_input BOG=data/BOG/BOG.Q \
  --external_condition_order AFR,EUR,AMR
```

For a combined BOG+BAQ model, both condition files are required:

```
--datasets '{BOG,BAQ}' \
--dataset_input BOG=data/BOG/genotypes.csv \
--dataset_input BAQ=data/BAQ/genotypes.csv \
--condition_input BOG=data/BOG/BOG.Q \
--condition_input BAQ=data/BAQ/BAQ.Q \
--condition_source external \
--bog_baq_harmonized
```

15.6 Direct SNP-space residual convolutional model

Start conservatively because memory use is much larger than for 50 PCs:

```
python conditional_wgan_pipeline.py ... \
  --model_backend genotype_conv \
  --batch_size 8 \
  --conv_channels 64 \
  --conv_blocks 4 \
  --pack_size 1
```

Then test `--pack_size2` and/or `--balanced_sampling` as separate ablations.

16 How to interpret outcomes

16.1 Do not optimize only PC overlap

PC1–PC2 can look convincing while locus-level or local-dependence statistics are wrong. Conversely, a direct SNP model can improve local statistics while slightly changing the PCA cloud. HAPNEST's evaluation explicitly treats MAF, population structure, LD, and nearest-neighbor adversarial accuracy as separate fidelity properties and reports trade-offs among metrics [Wharrie et al., 2023].

16.2 Recommended minimum acceptance table

For every run, compare at least:

a) MAF MAE and correlation;

b) heterozygosity mean and distribution;

c) LD/dosage-r^2 error versus physical distance if genomic positions are available;

d) PC-space MMD and PC alignment;

e) nearest-neighbor adversarial accuracy;

f) exact replay and membership-inference AUC using an independent holdout set.

The existing adjacent-SNP LD metric is useful for smoke tests but should be upgraded to distance-binned LD decay on non-LD-pruned SNPs for serious validation. LD-pruned training data intentionally removes much of the local correlation one would otherwise ask the model to reproduce.

17 What is proven here and what is empirical

It is important to separate mathematical facts from empirical hypotheses.

- **Proven/algebraic:** rank-k PCA reconstruction has covariance $U_k \Lambda_k U_k^T$; inverse-frequency sampling equalizes group sampling mass; conditional-independent binomial discretization preserves off-diagonal covariance in expectation but adds marginal variance; a residual block has Jacobian $I + J_F$; a missing mode appears in a real pack with probability $1 - (1 - \varepsilon)^m$.
- **Architectural rationale, not a theorem of better performance:** Conv1D, FiLM, residual blocks, and packing provide inductive biases or optimization paths that may help this dataset. They must be tested empirically.
- **Not a privacy guarantee:** low membership AUC, low nearest-neighbor leakage, or zero exact replay do not imply differential privacy.

18 Suggested experiment log

For each comparison, record:

1. exact git/file version or checksum of the code;
2. genotype and reference input hashes;

3. whether BOG/BAQ harmonization was completed and how alleles/strand were checked;
4. whether data were LD-pruned and, if so, the pruning threshold/window;
5. ancestry condition source and, for external conditions, ADMIXTURE K, reference panel, and column mapping;
6. backend, seed, validation fraction, balanced-sampling flag, pack size, and all model hyperparameters;
7. fidelity metrics on a fixed holdout set;
8. privacy diagnostics on train versus independent holdout individuals.

This is more valuable than selecting a single visually pleasing PCA plot because it makes results reproducible and scientifically interpretable.

19 Implementation status and limitations

The shared input QC report is now produced in each cWGAN and unconditional Yelmen training job, with the default Yelmen query imputation aligned to the cWGAN per-SNP mean rule for audited hard-call inputs and an explicit categorical hard-call exception for RBM/CRBM. The original three ancestry-conditioned training backends are retained, and a separate Yelmen (2023) unconditional WGAN, categorical-RBM/CRBM, and optional VAE benchmark is supplied. The 2023 scripts were exercised on small two-chromosome unphased fixtures, with WGAN 16K/65K shape checks, model checkpoint outputs, default evaluation without checkpoint loading, and explicit non-overwriting resampling. These tests do not establish publication-scale convergence or reproduction of the authors' reported statistics. In this revision the new paper-style evaluation was smoke-tested on small two-chromosome genotype matrices, a short baseline-PCA training produced checkpoint-specific method plots, archived checkpoint resampling/evaluation passed end to end, and a white-box critic-ranking smoke test ran on toy inputs. These tests do not establish performance on a full-scale cohort or GPU, and the mock attack train/test files are not privacy-valid independent cohorts.

The original `genotype_conv` backend (distinct from the new Yelmen WGAN) is an experimental implementation inspired by the genomic convolutional literature, not a line-for-line reproduction of Yelmen et al. [Yelmen et al., 2023]. In particular, it operates on unphased diploid dosage rather than phased 0/1 haplotypes, uses a simple learned locus-specific output bias plus a positional critic channel, and currently processes the full SNP vector rather than an explicitly windowed chromosome hierarchy. These differences should be stated whenever results are reported.

20 Summary

The most important conceptual change is to separate three distinct goals:

1. **Representation fidelity**: does the model preserve allele frequencies, genotype correlations/LD, heterozygosity, population structure, and higher-order dependence?
2. **Conditional fidelity**: does changing $c = (q_{AFR}, q_{EUR}, q_{AMR})$ change generated genotypes in the intended ancestry-consistent way?
3. **Privacy/generalization**: does the model generate new individuals rather than reproducing the training genomes?

The new unconditional Yelmen benchmarks test population-structure learning *without* externally estimated ancestry conditions, unlike the existing cWGAN. No single GAN loss or PCA figure answers all three. The modular code is designed so each proposed mechanism can be isolated, measured, and rejected if it does not improve the relevant holdout metrics.

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